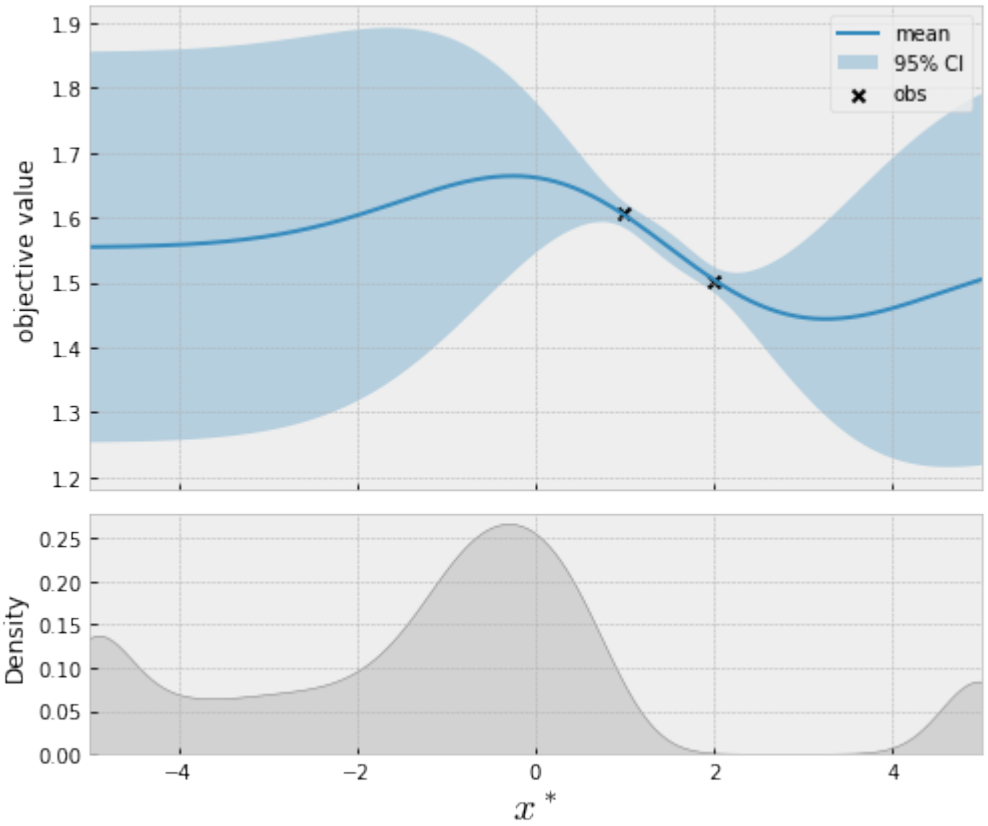


Figure 6.10 Gaussian process belief (top) and the distribution of the function optimizer x* (bottom). The distribution of the optimizer is non-Gaussian and fairly complicated, posing a challenge for modeling and decision making.

Making matters worse is the fact that we will need exponentially more samples to estimate the distribution of x^* when the dimensionality of the objective function increases.

IMPORTANT **The curse of dimensionality**

This is an instance of the curse of dimensionality in machine learning, which is often used to refer to the exponential cost of many procedures in machine learning with respect to the dimensionality of an object of interest.

In order to leverage entropy search in BayesOpt, we need to model our belief about the location of the optimum x^* using a probability distribution. However, we can't exactly model the distribution of the optimizer x^*; instead, we have to approximate it using samples drawn from a GP. Unfortunately, this process quickly becomes computationally expensive as the number of dimensions in the objective function increases.

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